Anthony J Barrows

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Education

PhD in Complex Systems and Data Science — 2022-2026 (Expected)
MS in Complex Systems and Data Science — 2019-2021
The University of Vermont, Burlington, Vermont

BA in Psychology — 2013–2016
University of Hartford, Hartford, Connecticut
Honors Thesis: *Characteristics Associated with MMPI-2 L-scale Elevations in a Sample of Police Officer Candidates*

Experience

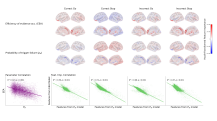
Teaching Fellow, ABCD-ReproNim Course — 2025-2026
Writing Consultant, Graduate Writing Center — 2023–Present
Program Representative, Graduate Student Senate — 2023–2024

Research Data Analyst, Vermont Center on Behavior and Health — 2021–2022
Lead Research Assistant, Vermont Center on Behavior and Health — 2019–2021
Research Assistant, Vermont Center on Behavior and Health — 2016–2019

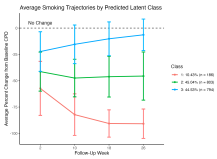
Awards

NEuroimaging Reserach VErmont (NERVE) Lab predoctoral fellow (T32 DA043593)
Harry L. Leonhard Award for Greatest Contribution to the Psychology Department by a Graduating Psychology Major

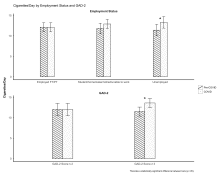
Publications



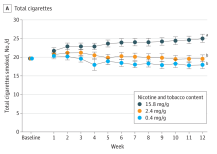
6. **Barrows, Anthony**, Weigard, A., McCabe, M., Potter, A., Garavan, H., and Allgaier, N. (2026). Cognitive Process Models Reveal Meaningful Brain–Behavior Associations in the Adolescent Brain Cognitive Development Study Stop-signal Task. *Journal of Cognitive Neuroscience*, pages 1–13



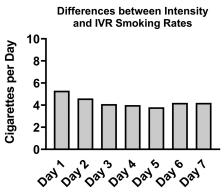
5. **Barrows, Anthony**, Klemperer, E., Garavan, H., Allgaier, N., Lindson, N., and Taylor, G. (2025). Smoking Reduction Trajectories and Their Association with Smoking Cessation: A Secondary Analysis of Longitudinal Clinical Trial Data. *BMJ Public Health*, 3(2):e001605



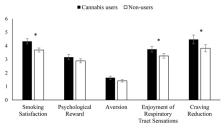
4. Wiley, R. C., Oliver, A. C., Snow, M. B., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D. C., Sigmon, S. C., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2021). The Impact of the Covid-19 Pandemic on Smoking Among Vulnerable Populations. *Nicotine & Tobacco Research*



3. Higgins, S. T., Tidey, J. W., Sigmon, S. C., Heil, S. H., Gaalema, D. E., Lee, D., Hughes, J. R., Villanti, A. C., Bunn, J. Y., Davis, D. R., Bergeria, C. L., Streck, J. M., Parker, M. A., Miller, M. E., DeSarno, M., Priest, J. S., Cioe, P., MacLeod, D., **Barrows, A. J.**, Markesich, C., and Harfmann, R. F. (2020). Changes in Cigarette Consumption With Reduced Nicotine Content Cigarettes Among Smokers With Psychiatric Conditions or Socioeconomic Disadvantage: 3 Randomized Clinical Trials. *JAMA Network Open*, 3(10):e2019311–e2019311



2. Nighbor, T. D., **Barrows, A. J.**, Bunn, J. Y., DeSarno, M. J., Oliver, A. C., Coleman, S. R. M., Davis, D. R., Streck, J. M., Reed, E. N., Reed, D. D., and Higgins, S. T. (2020). Comparing Participant Estimated Demand Intensity on the Cigarette Purchase Task to Consumption when Usual-Brand Cigarettes were Provided Free. *Preventive Medicine*



1. Parker, M. A., Streck, J. M., Bergeria, C. L., Bunn, J. Y., Gaalema, D. E., Davis, D. R., **Barrows, A. J.**, Sigmon, S. C., Tidey, J. W., Heil, S. H., and Higgins, S. T. (2018). Reduced Nicotine Content Cigarettes and Cannabis Use in Vulnerable Populations. *Tobacco Regulatory Science*, 4(5):94–91

Invited Talks

1. **Anthony Barrows** and Klemperer, E. (2026). Presentation of findings from Barrows et al. (2025) (*BMJ Public Health* 3(2)). Vermont Department of Health Tobacco Medicaid Benefit and Promotion Quarterly Meeting

Selected Talks

2. **Barrows, A. J.** (2023). Improving brain-behavior associations by strengthening models of behavior (Talk). *Adolescent Brain Cognitive Delopment (ABCD) Study 2023 Annual Meeting*

1. Taylor, S., Garavan, H., Weigard, A., Ging-Jehli, N., Hitchcock, P., and **Barrows, A. J.** (2023). What can computational modeling reveal about the generality or specificity of neurocognitive risk factors for psychopathology? (Panel Presentation). *American College of Neuropsychopharmacology 2023 Annual Meeting*

Abstracts

8. **Anthony Barrows**, David A. A. Baranger, S. A. S. O. N. A. and Garavan, H. (2026). Neural signatures help resolve individual differences using task fMRI in the abcd study. University of Vermont 2026 Annual Neuroscience, Research, and Behavior Forum (Poster Presentation)

7. **Anthony Barrows**, Hugh Garavan, A. P. N. A. M. A. (2025). Leveraging propensity score matching to test cannabis exposure effects in the abcd cohort. Flux Society Congress 2025, Dublin, Ireland (Poster Presentation)

6. **Anthony Barrows**, Klemperer, E., Garavan, H., Allgaier, N., Lindson, N., and Taylor, G. (2024). Smoking reduction trajectories and their association with smoking cessation. Society for Research on Nicotine and Tobacco 2024 Annual Meeting (Poster Presentation)

5. **Barrows, A. J.**, Pearson, E., Chaarani, B., Garavan, H., and Allgaier, N. (2023). Longitudinal task activation differences for the MID task in the ABCD Study (Poster Presentation). *Organization for Human Brain Mapping 2023 Annual Meeting*

4. Wiley, R. C., Oliver, A. C., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D., Sigmon, S., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2020a). Urban and rural differences in smoking behavior among vulnerable populations during the COVID-19 pandemic (Poster Presentation). *2020 Vermont Center on Behavior and Health Conference*

3. Wiley, R. C., Oliver, A. C., Snow, M. B., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D., Sigmon, S., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2020b). Smoking behavior among vulnerable populations in response to the COVID-19 pandemic (Abstract). *2020 Tobacco Centers of Regulatory Science Grantee Meeting*

2. Streck, J. M., Sigmon, S. C., Parker, M. A., Stitzer, M. L., Heil, S. H., Tidey, J. W., Gaalema, D. E., Bergeria, C. L., Davis, D. R., Priest, J., **Barrows, A. J.**, Rothman, M., Hughes, J. R., and Higgins, S. T. (2017). Subjective effects of smoking among opioid-maintained individuals: Results from a trial examining reduced nicotine content cigarettes (Abstract). *The College on Problems of Drug Dependence 2018 Annual Meeting*

1. **Barrows, A. J.** (2016). Personality characteristics associated with MMPI-2 L-Scale elevations (Poster Presentation). *2016 Eastern Psychological Association Annual Meeting*

Software Contributions

YabPlot A Python library for creating beautiful, publication-quality 3D brain visualizations.

VitroCal A Python library for analyzing data collected from *in vitro* calcium images.

Computer Skills

Programming Languages: R, Python, LaTeX, Markdown
Frameworks: R Shiny, R Markdown, Microsoft Access, Linux CLI, REDCap API
Developer Tools: R Studio, Jupyter, Visual Studio Code
Version Control: Git, GitHub
Libraries: Nilearn, tidyverse, ggplot2, lcmm, Plotly, pandas, NumPy, Matplotlib, scikit-learn
Software: Microsoft Excel, REDCap (Project Design)

References

[1] Higgins, S. T., Tidey, J. W., Sigmon, S. C., Heil, S. H., Gaalema, D. E., Lee, D., Hughes, J. R., Villanti, A. C., Bunn, J. Y., Davis, D. R., Bergeria, C. L., Streck, J. M., Parker, M. A., Miller, M. E., DeSarno, M., Priest, J. S., Cioe, P., MacLeod, D., **Barrows, A. J.**, Markesich, C., and Harfmann, R. F. (2020). Changes in Cigarette Consumption With Reduced Nicotine Content Cigarettes Among Smokers With Psychiatric Conditions or Socioeconomic Disadvantage: 3 Randomized Clinical Trials. *JAMA Network Open*, 3(10):e2019311–e2019311.

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[3] Parker, M. A., Streck, J. M., Bergeria, C. L., Bunn, J. Y., Gaalema, D. E., Davis, D. R., **Barrows, A. J.**, Sigmon, S. C., Tidey, J. W., Heil, S. H., and Higgins, S. T. (2018). Reduced Nicotine Content Cigarettes and Cannabis Use in Vulnerable Populations. *Tobacco Regulatory Science*, 4(5):94–91.

[4] Streck, J. M., Sigmon, S. C., Parker, M. A., Stitzer, M. L., Heil, S. H., Tidey, J. W., Gaalema, D. E., Bergeria, C. L., Davis, D. R., Priest, J., **Barrows, A. J.**, Rothman, M., Hughes, J. R., and Higgins, S. T. (2017). Subjective effects of smoking among opioid-maintained individuals: Results from a trial examining reduced nicotine content cigarettes (Abstract). *The College on Problems of Drug Dependence 2018 Annual Meeting*.

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[9] **Anthony Barrows**, David A. A. Baranger, S. A. S. O. N. A. and Garavan, H. (2026). Neural signatures help resolve individual differences using task fMRI in the abcd study. University of Vermont 2026 Annual Neuroscience, Research, and Behavior Forum (Poster Presentation).

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[17] Wiley, R. C., Oliver, A. C., Snow, M. B., Bunn, J. Y., **Barrows, A. J.**, Tidey, J. W., Lee, D. C., Sigmon, S. C., Gaalema, D. E., Heil, S. H., Markesich, C., Villanti, A. C., and Higgins, S. T. (2021). The Impact of the Covid-19 Pandemic on Smoking Among Vulnerable Populations. *Nicotine & Tobacco Research.*